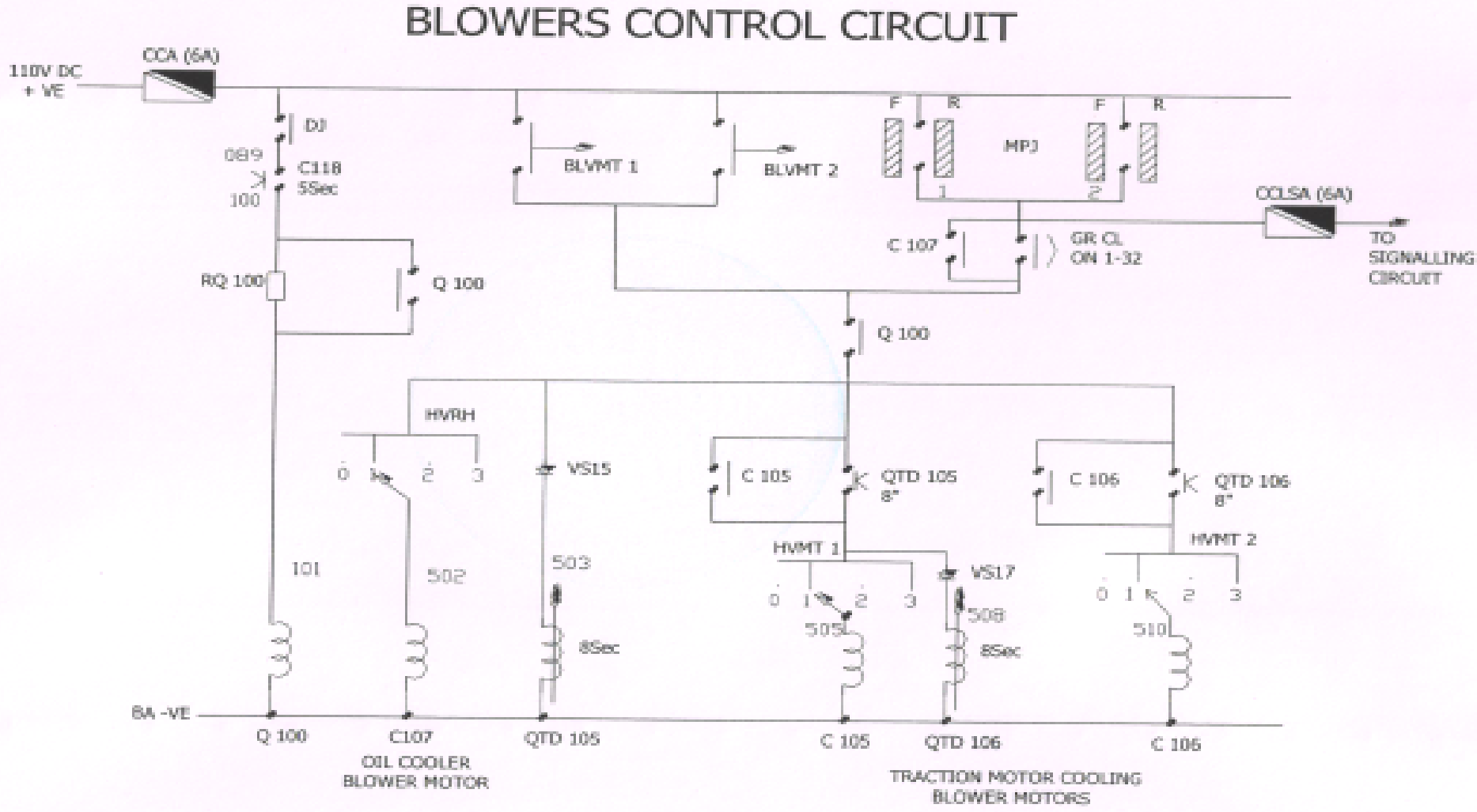
AUXILIARY CONTROL CIRCUIT ARNO FITTED LOCOMOTIVE

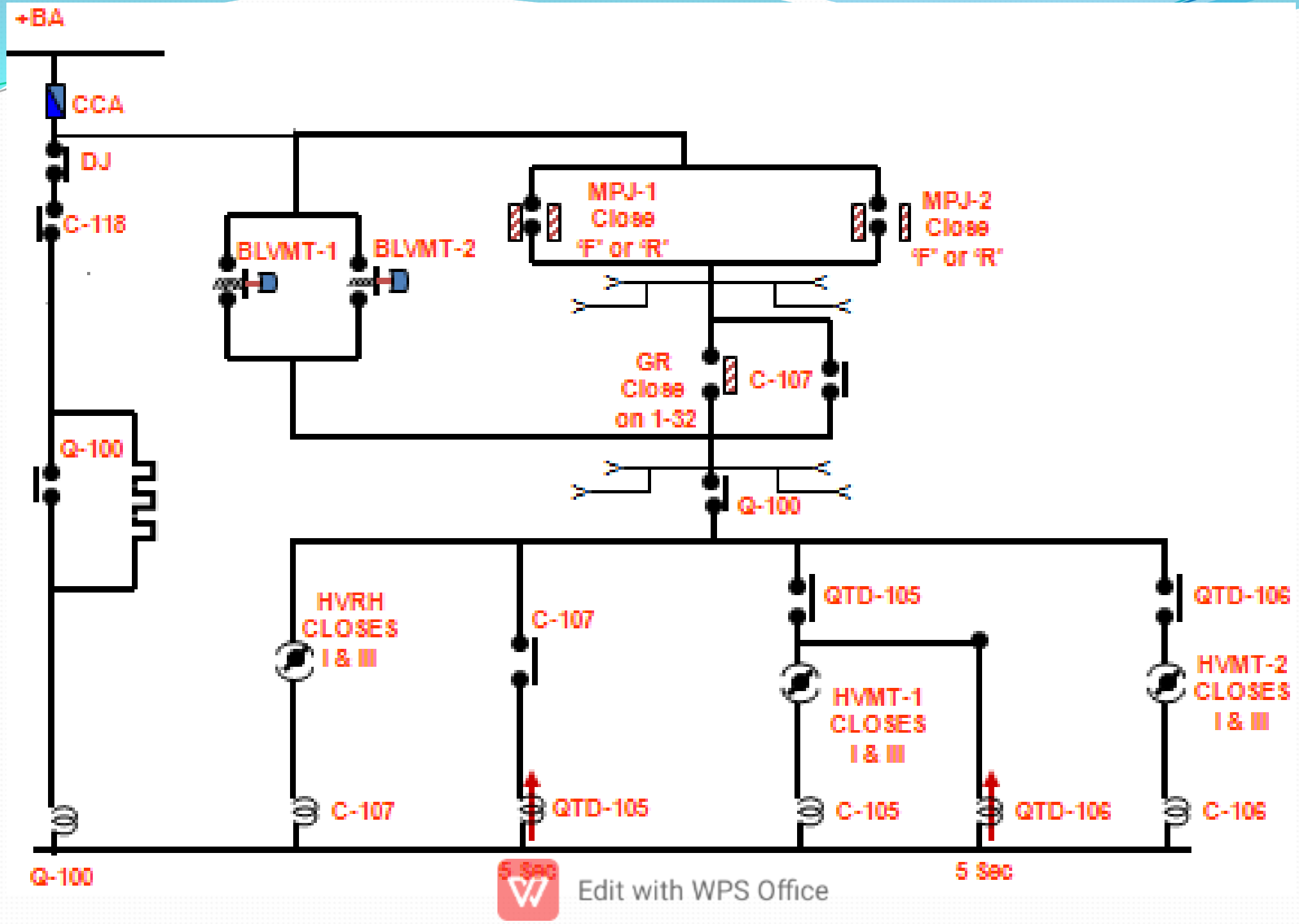
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BLOWER CONTROL CIRCUIT



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CONTROL CIRCUIT OF BLOWER CONTACTORS



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DESCRIPTION

- This circuit is intended to give 110 volts Battery voltage supply to the auxiliary motor contactor magnet coils.
- **Q100 Branch: -** This is auxiliary controlling relay.
- After closing DJ, this relay energizes through CCA fuse,
- DJ N/O I/L,
- C118 chronometric I/L(5 sec time delay).
- Q100 is energised,
- Q100 N/C self –I/L opens and path is maintained through RQ100.
- One N/O I/L of Q100 relay closes on compressors control circuit branch,
- second N/O I/L closes on blowers control circuit branch
- *If exhauster is provided third N/O I/L closes on exhausters control circuit branch

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CIRCUIT

After closing BLVMT 1 / 2 all blower contactor's coil get supply through Q100 N/O I/L.

First, C107 coil gets supply through HVRH switch closes on '1' or '3' position.

Along with C107, QTD 105 also energizes through C107 N/O interlock.

QTD105 relay's N/O I/L closes after 5 seconds on common path of C105, QTD106.

C105 contactor closes through HVMT1 closes on '1' of '3' position. After closing of C105 it's N/O I/L closes on its same path to avoid chattering of contactor,

Along with C105, QTD 106 also energizes.

QTD106 relay's N/O I/L closes after 5 seconds on C106 coil branch. Third, C106 contactor closes through QTD106 N/O I/L and HVMT2 close on '1'or '3' position.

After closing of C106 it's N/O I/L closes on its same path to avoid chattering of contactor.

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BLOWERS ARE NOT FUNCTIONING

- Operate BLVMT switch 2 - 3 times.
- If unsuccessful take one notch and check blower working or not.
- If unsuccessful wedge contactor(only one).

(If only 3 blower contactor not close , wedge C107 contactor.)

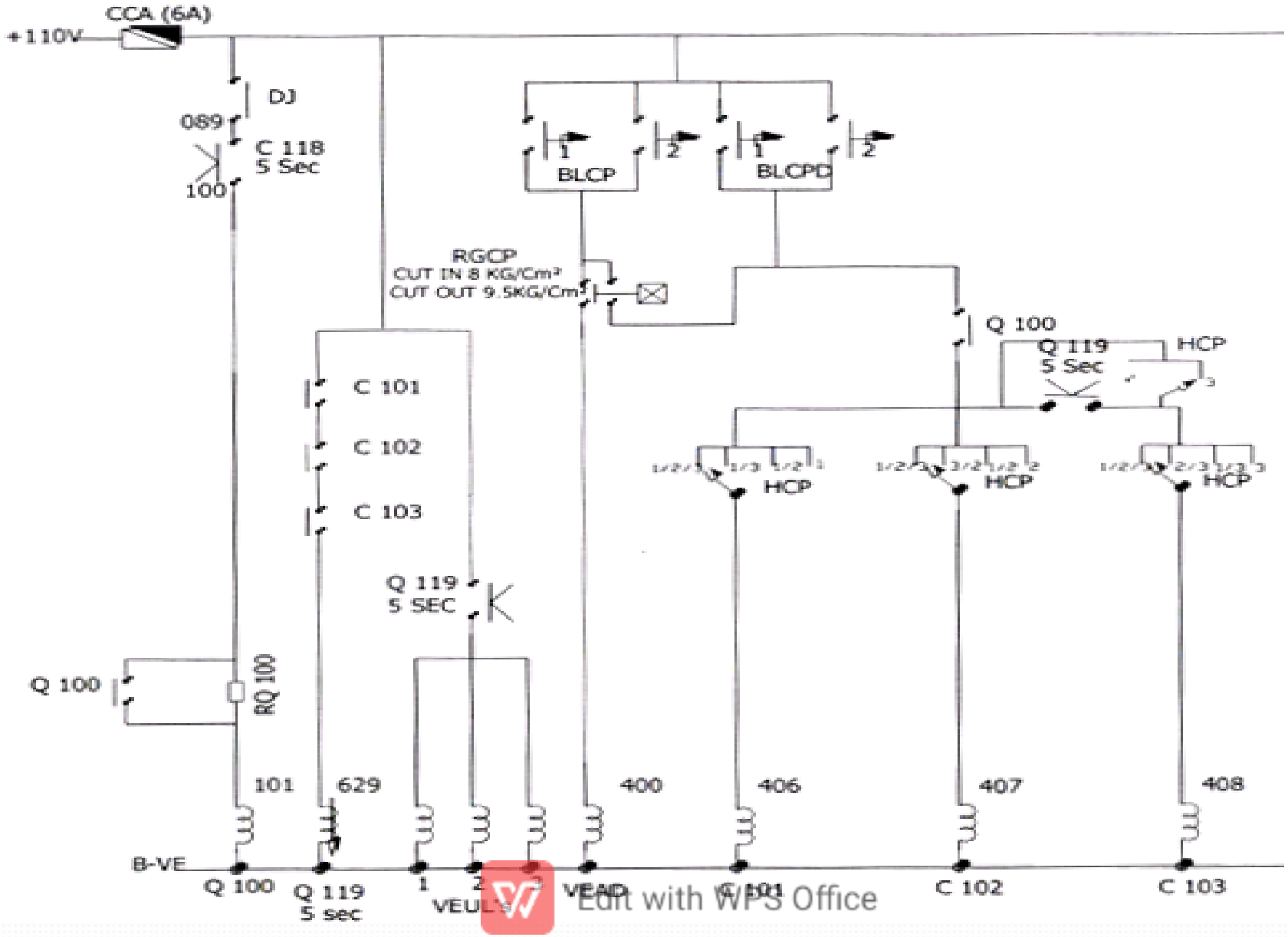
IF ANY PARTICULAR AUXILIARY CONTACTOR IS NOT CLOSING :-

Stop the train, open DJ, lower Pantograph, put 'OFF' HBA and wedge the contactor.

If Blower contactor wedge put blower switch on 3 rd position

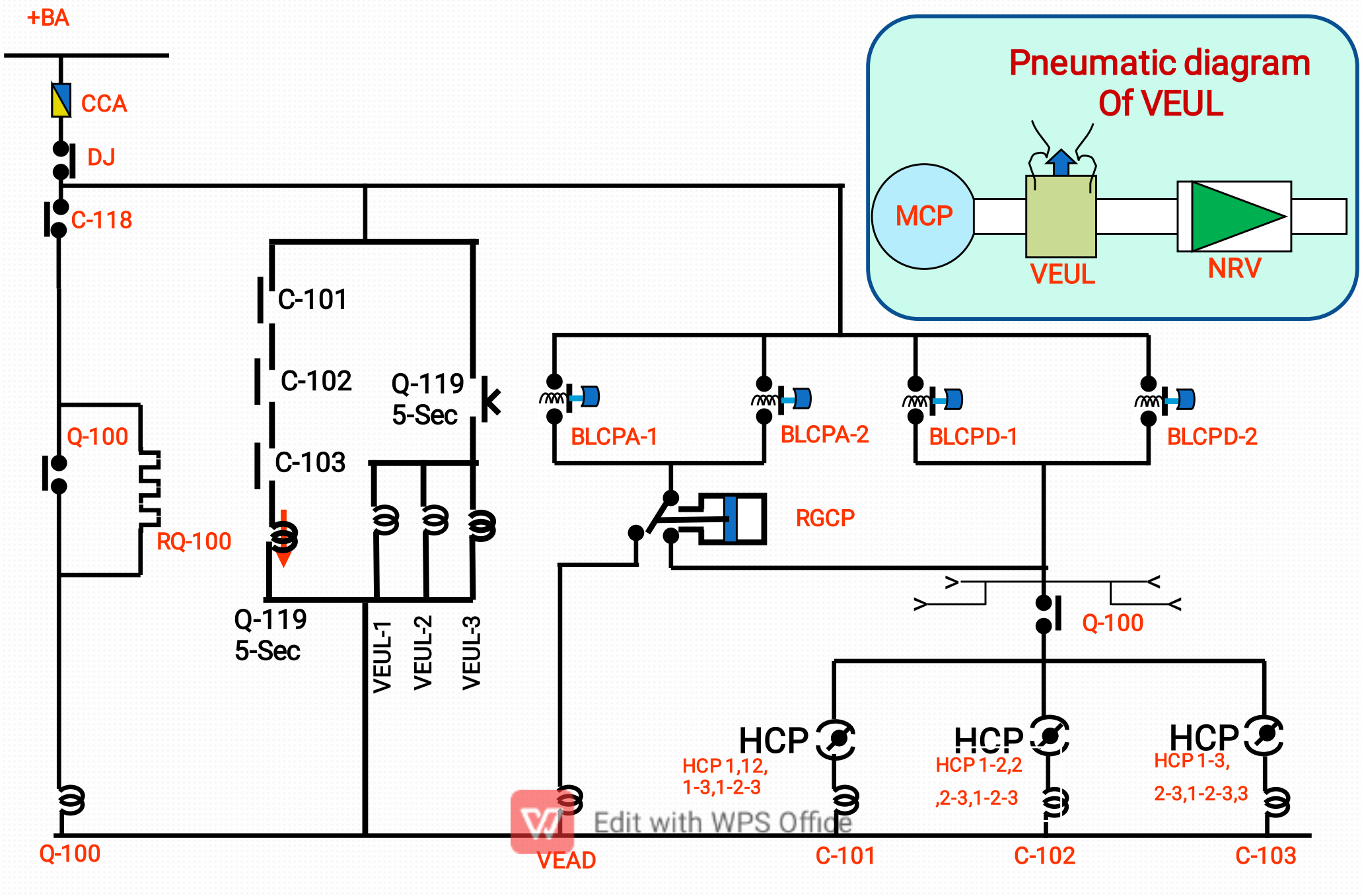
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COMPRESSORS CONTROL CIRCUIT



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CONTROL CIRCUIT OF MCP (Modified) With Q-119 & VEUL



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COMPRESSORS CONTROL CIRCUIT

After closing BLCP, compressor contactors coils gets supply according to HCP switch position through BLCP 1/2, RGCP I/L and Q 100 N/O I/L.

RGCP is provided to close the compressor contactors when MR pressure is 8Kg/cm^2 or less and open the compressor contactors when MR pressure is 9.5Kg/cm^2

When pressure reaches to 9.5Kg/cm^2, N/C I/L of RGCP opens on compressor contactors coil circuit and closes on VEAD magnet valve ckt to drain out moisture from MR 1 & 2 through ADV.

When compressor contactors are opened, closed N/C I/L of C101, C102 & C103 energize the relay Q119 . Q119 relay have 5sec time lag .

Un-loader valves energize to remove the back up pressure from each CP delivery pipe.

Isolating coc provided in un-loader valve to arrest leakage.

Q119 relay is provide for 5 min late start of last MCP after 5 seconds when we select 3 nos of MCP at atime.

To select required No. of MCPs, switch 'HCP' is provided. HCP switch is having 8 positions (0,1, 2, 3, 12,23 & 123).

Switch BLCPD is provided to direct start MCP without RGCP .

When MR pressure creates maximum up to 10.5 Kg/cm2 put off BLCPD.

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COMPRESSOR NOT WORKING

- Operate BLCPA switch 2 – 3 times.
- Put ‘ON’ BLCPD and try.
- Check HCP position (should not in ‘0’).
- Change HCP position and try.
- If unsuccessful wedge contactor (C101, C102, C103) as per requirement (But not more than two contactor) after take safety.

When wedge MCP contactor , MCP continuous working. Then open MR drain cock partially to reduce more air pressure in MR.

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ALL AUXILIARY NOT WORKING

NO Compressor and Blower contactors close (BLCP & BLVMT 'ON').

1. Check CCA fuse melt.
2. Check relay Q100 energize or not.

1. If CCA fuse melt :-

- Replace by new fuse.
- If CCA fuse melt again , replace by putting 'OFF' HOBA Switch.
- If CCA fuse melt again after putting 'OFF' HOBA Switch, then do not replace CCA fuse again and put HOBA switch 'ON' position.
- Take safety and wedge Auxiliary motor contactor as per requirement and inform TLC.
- Wedge 2 CP contactor for air brake train.
- Clear block section at 5 notch and ask for assist.

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ALL AUXILIARY NOT WORKING

2. **Check Q100 relay de-energize:-**

It should be remain energize when DJ closed.

- If found de-energize ,operate relay plunger 2 -3 times.
- If still de-energize, open and close the DJ 2 – 3 times.
- If still de-energize, open DJ ,put 'OFF' HBA and operate C118 contactor arm manually.
- If unsuccessful ,wedge Q100 relay in energize condition and work train, inform to TLC.

Note :- After DJ trip/Open put BLCP, BLVMT switch put 'OFF' position. These switch should be put 'ON' one by one after close DJ.

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